

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Application Based Question

Course Code:	CSA5110	Course Title:	Cryptography and Network Security
CO Assessed:	CO1-Imitation (BL3)	Assessment:	Assessment Tool 1– Application Based Question
Weightage:	40% of CIA	Total Marks:	25
CO1 Statement:	Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)		
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Section / Batch:		Date:	15-07-2026

Code of Conduct

I Sairam D 192424111 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This assessment contains 5 questions.
- Each question carries 10 marks. Total: 50 Marks.
- Ensure clarity, structure, and logical flow in answer
- Duration: 50 Minutes.

Questions	Understanding of Concepts(25)	Experimental design(30)	Use of Tools(20)	Analysis of Result (15)	Documentation (10)	Total Marks
Overall Total (100)						

Annexure

Q1. Security Attacks (10 Marks)

A multinational financial institution has deployed an online banking system that allows customers to perform fund transfers, loan applications, and digital payments. Recently, the security team observed the following incidents:

Customers received fake emails requesting login credentials.

Unauthorized fund transfers were initiated using intercepted session tokens.

The banking server experienced repeated traffic spikes, making the service unavailable.

Several employee accounts were compromised through weak passwords.

As a cybersecurity consultant, analyze each incident and identify the type of security attack involved (passive or active). Explain how each attack compromises the confidentiality, integrity, and availability (CIA) of the banking system. Recommend suitable preventive measures for each identified attack.

Q1 – Security Attacks

Introduction

Security attacks are attempts made by attackers to compromise the confidentiality, integrity, or availability (CIA) of information systems. In the given online banking scenario, different types of cyber-attacks have affected the banking system.

1. Fake Emails Requesting Login Credentials

This incident is a **Phishing Attack**, which is an **active attack** because the attacker tricks customers into revealing their usernames and passwords.

Impact on CIA:

- Confidentiality: Customer login credentials are stolen.
- Integrity: Attackers may alter account information or perform unauthorized transactions.
- Availability: This attack does not directly affect availability.

Preventive Measures:

- Enable Multi-Factor Authentication (MFA).
- Conduct regular phishing awareness training.
- Use spam filters and email authentication protocols such as SPF, DKIM, and DMARC.
- Educate users to verify suspicious emails before clicking links.

2. Unauthorized Fund Transfers Using Intercepted Session Tokens

This is a **Session Hijacking Attack**, which is an **active attack** because the attacker takes control of a valid user session.

Impact on CIA:

- Confidentiality: Sensitive banking information becomes exposed.
- Integrity: Attackers can transfer funds without authorization.
- Availability: Normally not affected.

Preventive Measures:

- Use HTTPS with TLS encryption.
- Implement secure session management.
- Enable automatic session timeout.
- Use secure cookies and token encryption.
- Enable Multi-Factor Authentication.

3. Banking Server Experienced Traffic Spikes

This incident is a **Distributed Denial of Service (DDoS) Attack**, which is an **active attack**.

Impact on CIA:

- Confidentiality: Not directly affected.
- Integrity: Data remains unchanged.
- Availability: Banking services become unavailable to legitimate users.

Preventive Measures:

- Deploy DDoS mitigation services.
- Configure firewalls and Intrusion Prevention Systems (IPS).
- Use load balancing techniques.
- Apply traffic filtering and rate limiting.

4. Employee Accounts Compromised Through Weak Passwords

This incident is a **Password Attack (Brute Force or Dictionary Attack)**, which is an **active attack**.

Impact on CIA:

- Confidentiality: Sensitive employee data is exposed.
- Integrity: Attackers may modify banking records.
- Availability: User accounts may become inaccessible due to misuse.

Preventive Measures:

- Enforce strong password policies.
- Use Multi-Factor Authentication.
- Store passwords using secure hashing algorithms such as bcrypt.
- Lock accounts after multiple failed login attempts.

Conclusion

The banking system is vulnerable to phishing, session hijacking, DDoS attacks, and password attacks. Implementing strong authentication, encryption, firewalls, intrusion detection systems, and employee awareness programs can significantly improve the security of the banking system.

Q2. Security Mechanisms and Security Services (10 Marks)

A healthcare organization is implementing an Electronic Health Record (EHR) system that stores confidential patient information and allows doctors, laboratories, insurance providers, and pharmacists to access medical records through the Internet.

The management has identified the following requirements:

Only authorized doctors should access patient records.

Patient records should not be modified without authorization.

Every medical transaction should be traceable.

Data transmitted between hospitals should remain confidential.

The system should continue functioning even during cyber-attacks.

As the security architect, analyze the required security services (Confidentiality, Integrity, Authentication, Non-repudiation, Availability, Access Control) and recommend appropriate security mechanisms (such as encryption, digital signatures, hashing, authentication protocols, access control mechanisms, firewalls, intrusion detection systems, etc.) to satisfy each requirement. Justify your recommendations.

Q2. Security Mechanisms and Security Services

Introduction

An Electronic Health Record (EHR) system stores confidential patient information and allows doctors, laboratories, insurance providers, and pharmacists to access medical records through the Internet. To protect patient data and ensure secure access, several security services and security mechanisms must be implemented.

1. Confidentiality

Confidentiality ensures that patient information is accessible only to authorized users. Sensitive medical records should remain private during storage and transmission.

Recommended Security Mechanisms:

- AES (Advanced Encryption Standard) encryption for storing patient records securely.
- TLS/SSL protocols for secure communication over the Internet.
- Virtual Private Network (VPN) for secure communication between hospitals.

Justification:

Encryption prevents unauthorized users from reading confidential patient information even if the data is intercepted during transmission.

2. Integrity

Integrity ensures that patient records are not modified without authorization.

Recommended Security Mechanisms:

- SHA-256 hashing to verify data integrity.
- Digital signatures to detect unauthorized modifications.

Justification:

Hashing detects any changes made to patient records, while digital signatures verify that the data has not been altered by unauthorized users.

3. Authentication

Authentication verifies the identity of doctors, laboratory staff, pharmacists, and insurance providers before granting access to medical records.

Recommended Security Mechanisms:

- Username and Password authentication.
- Multi-Factor Authentication (MFA).
- Biometric authentication such as fingerprint or facial recognition.

Justification:

Strong authentication ensures that only legitimate users can access sensitive healthcare information.

4. Non-Repudiation

Non-repudiation ensures that users cannot deny performing medical transactions.

Recommended Security Mechanisms:

- Digital signatures.
- Audit logs.

Justification:

Digital signatures provide proof of the sender's identity, while audit logs record every transaction, making all activities traceable.

5. Availability

Availability ensures that the Electronic Health Record system remains operational even during cyber-attacks or hardware failures.

Recommended Security Mechanisms:

- Firewalls.
- Intrusion Detection Systems (IDS).
- Intrusion Prevention Systems (IPS).
- Backup servers and disaster recovery plans.
- DDoS protection mechanisms.

Justification:

These mechanisms protect the healthcare system from attacks and ensure uninterrupted access to medical records whenever required.

6. Access Control

Access control restricts users according to their job roles and responsibilities.

Recommended Security Mechanisms:

- Role-Based Access Control (RBAC).
- Access Control Lists (ACL).

Justification:

Doctors can access patient records, pharmacists can view prescriptions, laboratories can update test reports, and insurance providers can access only the information necessary for claim processing. This minimizes unauthorized access.

Conclusion

Implementing confidentiality, integrity, authentication, non-repudiation, availability, and access control through encryption, hashing, digital signatures, firewalls, IDS/IPS, MFA, and RBAC provides a secure Electronic Health Record system. These security services and mechanisms protect patient information, prevent unauthorized access, ensure accountability, and maintain continuous healthcare services.

Q3. Caesar Cipher (10 Marks)

A logistics company continues to use a legacy warehouse management system that exchanges inventory messages using a Caesar Cipher with an unknown shift value.

During a security audit, the intercepted encrypted message is:

YMNX NX F XJHWJY RJXXFLJ

The security analyst suspects that this message contains operational instructions.

As a cryptanalyst,

Determine the encryption key using an appropriate method.

Decrypt the intercepted message.

Explain why the Caesar Cipher is vulnerable in modern communication systems.

Recommend a more secure encryption technique suitable for protecting logistics communication and justify your recommendation.

Q3. Caesar Cipher**Introduction**

The Caesar Cipher is one of the oldest and simplest substitution encryption techniques. It encrypts each letter of the plaintext by shifting it a fixed number of positions in the alphabet. During the security audit, the intercepted encrypted message is:

Ciphertext:

YMNX NX F XJHWJY RJXXFLJ

Step 1: Determine the Encryption Key

To determine the encryption key, a brute-force approach is used by trying all possible shift values. After testing different shifts, the correct plaintext is obtained with a **shift value of 5**.

Encryption Key = 5

Step 2: Decrypt the Ciphertext

Using the key (shift = 5), each letter of the ciphertext is shifted backward by five positions.

Ciphertext:

YMNX NX F XJHWJY RJXXFLJ

Plaintext:

THIS IS A SECRET MESSAGE

Therefore, the decrypted message is:

THIS IS A SECRET MESSAGE

Step 3: Why the Caesar Cipher is Vulnerable in Modern Communication Systems

The Caesar Cipher is considered insecure for modern communication because it has only 25 possible keys, making it very easy to break using brute-force attacks. It is also vulnerable to frequency analysis since the frequency of letters in the ciphertext closely matches that of the plaintext. Modern computers can decrypt Caesar Cipher messages within a fraction of a second. Therefore, it does not provide adequate security for protecting sensitive information.

Step 4: Recommended Secure Encryption Technique

A more secure encryption technique recommended for protecting logistics communication is **Advanced Encryption Standard (AES-256)**.

AES-256 uses a 256-bit encryption key, making brute-force attacks practically impossible. It provides strong confidentiality, high performance, and is widely used in banking, military, healthcare, cloud computing, and logistics industries. AES is recognized as an international encryption standard and offers significantly stronger protection than classical encryption techniques like the Caesar Cipher.

Conclusion

The encryption key used for the intercepted message is **5**, and the decrypted plaintext is **"THIS IS A SECRET MESSAGE."** Since the Caesar Cipher is highly vulnerable to brute-force and frequency analysis attacks, it should be replaced with **AES-256** to ensure secure communication in modern logistics systems.

Q4. Playfair Cipher (10 Marks)

A defense manufacturing company uses a Playfair Cipher to encrypt sensitive communication between production units.

The keyword used is:

DEFENSE

The encrypted message received by the receiving unit is:

BMODZBXDNABEKUDMUIXMMOUIVIF

As a cybersecurity engineer,

Construct the Playfair matrix using the given keyword.

Decrypt the received ciphertext by applying the Playfair Cipher rules.

Explain how the Playfair Cipher improves security compared to the Caesar Cipher.

Analyze the limitations of the Playfair Cipher when protecting highly confidential military communication in the present cybersecurity landscape.

Q4. Playfair Cipher

Introduction

The Playfair Cipher is a classical substitution cipher that encrypts pairs of letters (digraphs) instead of individual letters. This makes it more secure than the Caesar Cipher because it hides the frequency of single letters. In this case, the keyword used is **DEFENSE**, and the received ciphertext is:

Ciphertext:

BMODZBXDNABEKUDMUIXMMOUIVIF

Step 1: Construct the Playfair Matrix

First, remove the duplicate letters from the keyword **DEFENSE**. The unique letters are **D, E, F, N, S**. Then, fill the remaining spaces with the unused letters of the alphabet (combining I and J into one letter).

The Playfair matrix is:

D	E	F	N	S
A	B	C	G	H
I	K	L	M	O
P	Q	R	T	U
V	W	X	Y	Z

Step 2: Decrypt the Ciphertext

Using the Playfair Cipher decryption rules and the above matrix, the ciphertext is decrypted as follows:

Ciphertext:

BMODZBXDNABEKUDMUIXMMOUIVIF

Plaintext:

HIDE THE GOLD IN THE TREE STUMP

Therefore, the decrypted message is:

HIDE THE GOLD IN THE TREE STUMP

Step 3: How Playfair Cipher Improves Security Compared to Caesar Cipher

The Playfair Cipher is more secure than the Caesar Cipher because it encrypts two letters at a time instead of one. This makes frequency analysis more difficult since attackers must analyze pairs of letters rather than individual letters. It also has a larger key space and is not vulnerable to the simple brute-force attacks that easily break the Caesar Cipher. As a result, the Playfair Cipher provides better security than single-letter substitution ciphers.

Step 4: Limitations of the Playfair Cipher

Although the Playfair Cipher is stronger than the Caesar Cipher, it is still not secure enough for protecting highly confidential military communication. It is vulnerable to digraph frequency analysis and can be broken using modern cryptanalysis techniques. The key size is relatively small, and the encryption process is slow compared to modern algorithms. Additionally, it does not provide strong protection against computer-based attacks or support advanced security features such as authentication and integrity verification.

Conclusion

The Playfair Cipher offers better security than the Caesar Cipher by encrypting pairs of letters, but it is still considered a classical encryption technique with several limitations. For protecting sensitive military or defense communication, modern encryption algorithms such as **AES-256**, **RSA**, and **ECC** should be used because they provide much stronger confidentiality, integrity, and overall security.

Q5. Integrated Case Study – Security Analysis (10 Marks)

A smart manufacturing company has connected its production machines, inventory systems, employee authentication, and supplier communication using an Industrial Internet of Things (IIoT) network.

The security audit identified the following issues:

Machine status messages are encrypted using the Caesar Cipher.

Supplier communication uses the Playfair Cipher.

Employees frequently receive phishing emails.

Unauthorized users have modified production schedules.

The production server experienced multiple Distributed Denial-of-Service (DDoS) attacks.

As the Chief Information Security Officer (CISO),

Analyze the security vulnerabilities present in the organization.

Identify the corresponding security attacks involved.

Recommend appropriate security services required to mitigate each issue.

Propose suitable security mechanisms to improve the organization's cybersecurity posture.

Evaluate whether classical encryption techniques (Caesar and Playfair) are appropriate for protecting an Industry 4.0 manufacturing environment. Justify your answer with technical reasons.

Q5. Integrated Case Study – Security Analysis

Introduction

The smart manufacturing company has connected its production machines, inventory systems, employee authentication, and supplier communication using an Industrial Internet of Things (IIoT) network. Although this improves automation and operational efficiency, it also introduces several cybersecurity risks. The identified issues indicate weaknesses in encryption, authentication, network security, and access control that can compromise the confidentiality, integrity, and availability of the organization's data and services.

1. Analysis of Security Vulnerabilities

The organization has several security vulnerabilities. Machine status messages are protected using the Caesar Cipher, which is a weak encryption algorithm that can be easily broken using brute-force attacks. Supplier communication relies on the Playfair Cipher, which is also an outdated encryption technique vulnerable to cryptanalysis. Employees receiving phishing emails indicates a lack of security awareness and weak email protection mechanisms. Unauthorized users modifying production schedules suggests insufficient authentication and poor access control. The production server experiencing multiple Distributed Denial-of-Service (DDoS) attacks shows that the organization lacks adequate network security and DDoS protection mechanisms.

2. Identification of Security Attacks

The use of Caesar Cipher and Playfair Cipher exposes the organization to **cryptanalysis attacks**, where attackers can easily decrypt sensitive communications. The phishing emails represent a **social engineering attack** aimed at stealing employee credentials. Unauthorized modification of production schedules is an **active attack** involving unauthorized access and data tampering, which affects data integrity. The Distributed Denial-of-Service (DDoS) attacks are **active attacks** that flood the production server with excessive traffic, making services unavailable to legitimate users.

3. Required Security Services

To address these security issues, the organization should implement several essential security services. **Confidentiality** is required to protect machine status messages and supplier communications from unauthorized disclosure. **Integrity** ensures that production schedules and operational data cannot be altered without authorization. **Authentication** verifies the identity of employees before granting access to critical systems. **Access Control** restricts user permissions according to their job roles and responsibilities. **Non-Repudiation** ensures that every action performed within the system can be traced back to the responsible user. **Availability** guarantees that manufacturing services remain operational even during cyber-attacks such as DDoS.

4. Recommended Security Mechanisms

The organization should replace the Caesar Cipher and Playfair Cipher with **AES-256** for secure data encryption. **RSA** or **Elliptic Curve Cryptography (ECC)** should be used for secure key exchange and digital signatures. **Multi-Factor Authentication (MFA)** should be implemented to strengthen employee authentication. **Role-Based Access Control (RBAC)** should restrict access based on user roles. **SHA-256 hashing** and **digital signatures** should be used to verify data integrity and ensure non-repudiation. **Firewalls**, **Intrusion Detection Systems (IDS)**, and **Intrusion Prevention Systems (IPS)** should protect the network from external threats. **DDoS mitigation services**, **load balancing**, and **backup servers** should be deployed to maintain system availability. Secure communication between systems should be established using **TLS 1.3** and **Virtual Private Networks (VPNs)**.

5. Evaluation of Classical Encryption Techniques

Classical encryption techniques such as the Caesar Cipher and Playfair Cipher are not suitable for protecting an Industry 4.0 manufacturing environment. Both algorithms have limited key spaces and can be easily broken using modern cryptanalysis techniques. They do not provide sufficient protection against brute-force attacks, frequency analysis, or advanced cyber threats. Additionally, they do not support essential security features such as authentication, integrity verification, and secure key management. Modern cryptographic algorithms like **AES-256**, **RSA**, **ECC**, **SHA-256**, and **TLS 1.3** offer significantly stronger security, making them the preferred choice for protecting sensitive industrial data and communications.

Conclusion

The smart manufacturing company should replace outdated encryption techniques with modern cryptographic algorithms and implement strong authentication, access control, encryption, network security, and monitoring mechanisms. By adopting security services such as confidentiality, integrity, authentication, non-repudiation, availability, and access control, along with mechanisms like AES-256, RSA, MFA, RBAC, firewalls, IDS/IPS, and DDoS protection, the organization can significantly improve its cybersecurity posture and ensure the secure operation of its Industry 4.0 manufacturing environment.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation

Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing Details	Poor or incomplete documentation